

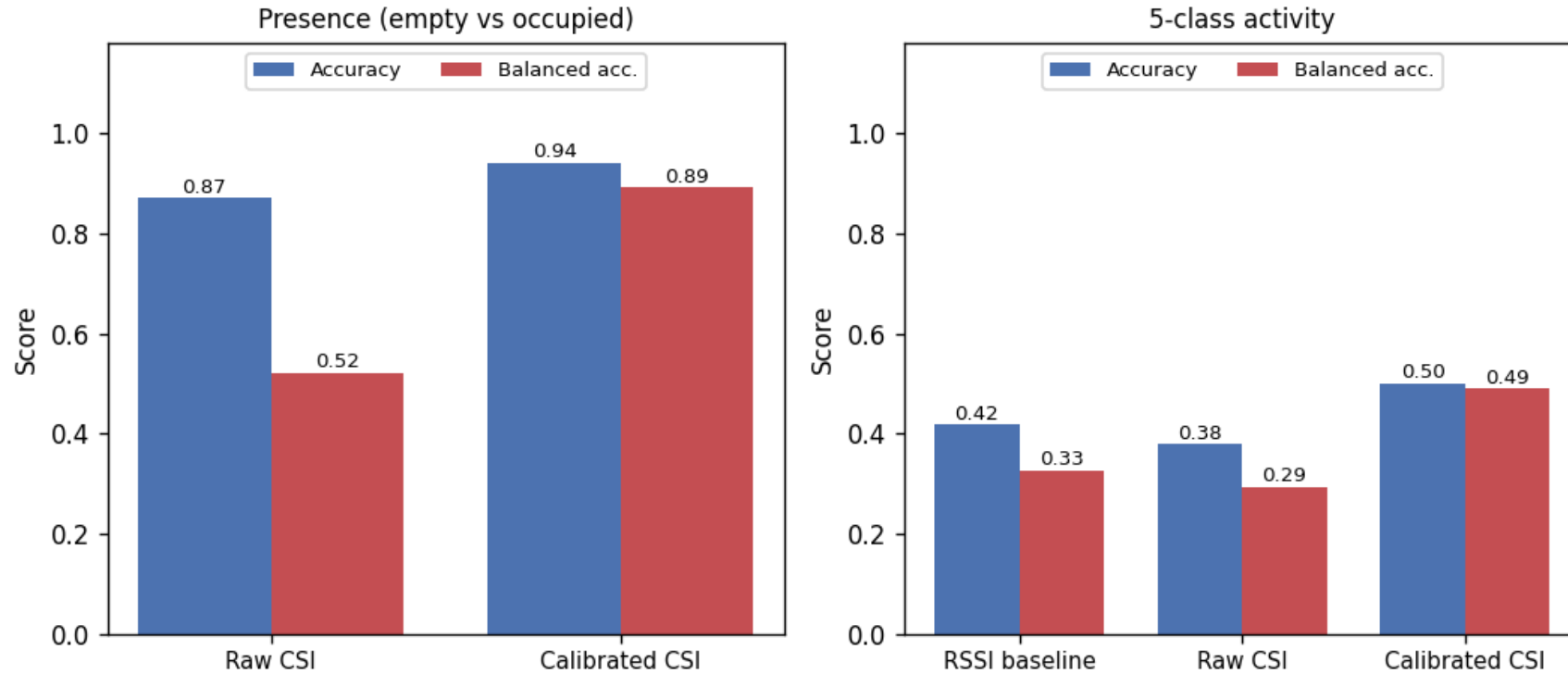
Wi-Fi CSI Presence & Activity Sensing

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Setup

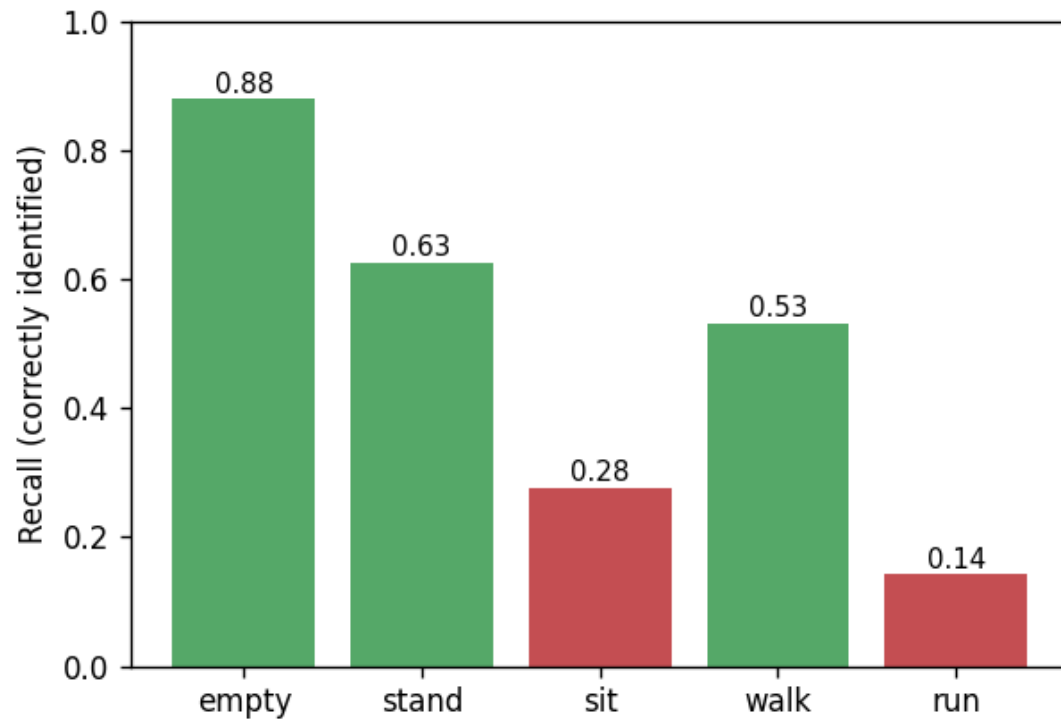
- Library study room. Five configurations = different board placements and antenna orientations
- Each configuration runs the same script: empty / stand / sit / walk / run.
- Calibrated against the room's own empty baseline. Tested on a configuration never seen in training

Presence Accuracy



- Presence on an unseen configuration: 94% accuracy, 0.89 balanced.
- Without calibration the model just answers “occupied” (0.52 balanced)
- Activity (5 classes) is harder at ~0.50 which is still ahead of RSSI-only (0.42) and raw CSI (0.38).

Classification



- Empty, standing and walking are recognizable (0.53–0.88).
- Sitting (0.28) gets confused with standing and running (0.14) gets confused with walking.

Calibration detail: the empty baseline

